Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

Structure-preserving representation before decision contraction.

`pip install erisml-compiler` · `github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc`

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PyTorch Conference North America 2026 — Responsible AI poster

1

Don't collapse to a scalar.

the moral tensor (who × what is owed)

✓ keep the structure

who is affected · what is owed
who consented · who bears imposed risk

contract to a decision only at the end, and log the contraction:
weights · who lost · what residue remains

2

pip install erisml-compiler — structure before the scalar

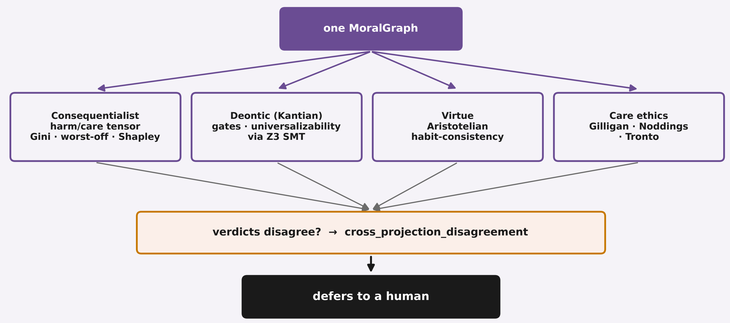
MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

3

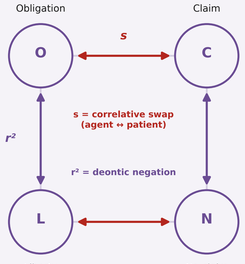
Four lenses. No silent winner.



"We run four and show you where they conflict — we don't pick one and hide it."

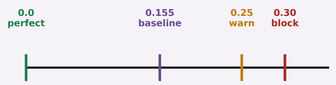
4

Is this 'good'? → Does it preserve the bonds?



s and r² commute → V4 (Klein four-group, order 4) — measured

full D4 (order 8) posited, now tested: a 09/26 hunt in a small LM found no quarter-turn (non-abelian signature absent). V4 stands.



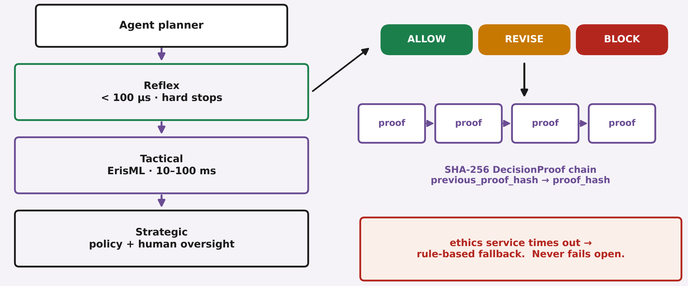
Bond Index: does the judgment survive the swap?

multi-rank tensors: party × time × action × coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

5

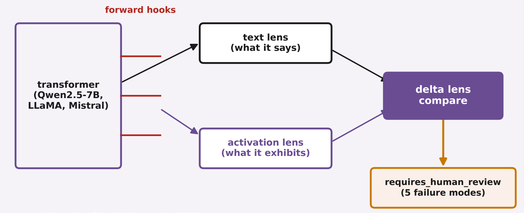
Safety in the loop. Fails safe.



between the agent's planner and its actuators

6

What the model says vs what it exhibits



Activation lens: first calibration numbers (09/26). Consumer-metric gating catches 19-21 pp more real behavior changes than raw L2 at matched flag rates. Replicated 2/2: second model + second transform. Probe mid-layers: the final-layer β map fails held-out validation in all three runs.

turboquant-pro (PyPI): PyTorch-native compression
HF 1-liner: TurboQuantCache() · Triton + Volta kernels · vLLM ~5× KV
live here: 3-bit codec on NATS · claims CI-gated · measured false-clear diag (v2.0.0a3)

7

Auditable means you can replay the judgment, and the regime that made it

full_response	NORMAL	EMERGENCY
Margaret	0.185 forbid	0.705 prefer
Daughter	0.402 neutral	0.902 prefer
EMS crew	0.318 forbid	0.902 prefer
Agency	0.200 forbid	0.802 prefer

decision: stand_by (0.834) → full_response (0.804); stand_by inverts to 0.307

elevation gate: detector 0.97 ≥ 0.90 → GRANTED
scope: paramedics only. Auto-reverts. Both edges audited.

conf 0.89 → REFUSED. Fallback: human review plus the EMS call, with no record share. A spurious emergency unlocks nothing.

worst-off shifts Daughter → Margaret (scoped privacy cost)
Shapley authority: Margaret-led → EMS-led

Three DecisionProofs: normal, emergency, and the elevation edge

Context elevation is privilege escalation. The elevation is authenticated, bounded, least-privilege, and audited. examples/care_robot_regimes reproduces every number.

One command. Real numbers. A hash you can verify.

```
$ python -m erisml.examples.care_robot_regimes
```

Same action, two regimes: forbid → obligated. Exact Shapley.
Authenticated elevation, audited on both edges. Replay the judgment.

1. Structure before contraction — keep the tensor, log the collapse + the residue.
2. Pluralism is the responsible move — four lenses, defer to a human on conflict.
3. Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early; first calibration 09/26 (consumer-metric gating, replicated 2/2)
ranks 1–3 real, higher partial · V4 measured, D4 posited (Lean + Mathlib verified)
embodied = design target · conversational agents = live

erismel-compiler **erisml-lib** **agi-hpc**